sMDD (k = ceil(N/2), test: connection_index) vs MDD — DD Python, paired rows (same instance, DD type, width and reduction)
MDD: results_python_01_11_2026_17-48-09.csv | sMDD: results_python_semi_semi_connection_index_03_11_2026_14-32-27.csv
Columns are means over the paired rows; 'ratio' is the geometric mean of sMDD/MDD per row. Time = construction + reduction; Solve = longest/shortest path over the built diagram.
Bound means skip rows where either value is ±inf (empty restricted / unreachable terminal), counted in 'non-fin.'. 'sMDD +/=/-': rows where the sMDD bound is better / equal / worse.

Problem	DD	Red.	N	Time MDD	Time sMDD	ratio	Solve MDD	Solve sMDD	ratio	Nodes MDD	Nodes sMDD	ratio	Arcs MDD	Arcs sMDD	ratio	Bound MDD	Bound sMDD	non-fin.	sMDD +/=/-
Knapsack	Exact	no	21	7.55	14.45	1.85×	2.67	3.32	1.48×	906,750	877,435	0.54×	1.69e+06	5.54e+06	2.25×	7,599	7,599	0	0/21/0
	Exact	yes	21	9.49	20.68	2.17×	2.55	4.59	1.90×	872,545	872,545	1.00×	1.64e+06	5.51e+06	2.71×	7,599	7,599	0	0/21/0
	Restricted	no	156	13.26	18.87	1.65×	4.34	5.22	1.43×	1.81e+06	1.78e+06	0.61×	3.05e+06	5.41e+06	1.81×	15,583	15,583	0	0/156/0
	Restricted	yes	156	15.83	24.55	1.87×	4.22	5.48	1.63×	1.77e+06	1.77e+06	0.97×	3e+06	5.39e+06	2.05×	15,583	15,583	0	0/156/0
	RelaxPriority	no	156	15.49	17.62	1.44×	4.12	4.45	1.33×	1.81e+06	1.79e+06	0.62×	3.06e+06	4.67e+06	1.65×	591,205	590,478	0	55/101/0
	RelaxPriority	yes	156	18.27	22.49	1.59×	4.15	5.04	1.44×	1.77e+06	1.75e+06	0.95×	3e+06	4.59e+06	1.80×	591,205	590,478	0	55/101/0
Independent Set	Exact	no	13	1.15	3.04	1.56×	0.2992	0.2397	0.80×	183,180	163,613	0.66×	213,062	503,285	1.18×	29.08	29.08	0	0/13/0
	Exact	yes	13	1.34	3.62	1.52×	0.3014	0.3406	0.82×	183,180	163,613	0.66×	213,062	503,285	1.18×	29.08	29.08	0	0/13/0
	Restricted	no	363	11.22	31.79	2.37×	1.52	1.69	1.07×	1.04e+06	846,781	0.80×	1.09e+06	1.24e+06	1.05×	26.93	22.15	0	0/121/242
	Restricted	yes	362	12.06	32.83	2.29×	1.47	1.85	1.06×	1.02e+06	812,049	0.77×	1.07e+06	1.15e+06	0.98×	26.95	22.17	0	0/121/241
	RelaxPriority	no	362	18.64	29.89	1.43×	2.47	2.70	1.07×	1.72e+06	1.77e+06	0.96×	1.83e+06	2.21e+06	1.14×	294	310	0	108/61/193
	RelaxPriority	yes	359	20.27	31.00	1.39×	2.31	0.9186	0.45×	1.38e+06	527,307	0.43×	1.48e+06	625,252	0.49×	292	309	0	105/61/193
Set Cover	Exact	no	178	0.3525	4.58	2.40×	0.0527	1.16	3.96×	39,335	34,222	0.99×	76,791	2.29e+06	6.16×	3.61	3.61	0	0/178/0
	Exact	yes	178	0.3992	7.85	3.29×	0.0438	0.4574	3.57×	33,371	32,017	1.00×	65,707	1.24e+06	6.10×	3.61	3.61	0	0/178/0
	Restricted	no	2048	8.98	10.55	1.93×	0.3723	1.75	3.18×	220,623	265,899	1.10×	425,912	3.19e+06	4.00×	8.78	8.78	0	3/2042/3
	Restricted	yes	2036	9.43	13.02	2.39×	0.3332	0.7943	2.44×	193,143	187,110	1.02×	376,383	1.47e+06	3.20×	8.85	8.85	0	3/2030/3
	RelaxPriority	no	2028	7.90	22.62	2.75×	0.3214	1.55	3.21×	189,849	245,364	1.16×	375,681	3.04e+06	4.27×	2.55	2.48	0	16/1851/161
	RelaxPriority	yes	2012	7.93	24.39	3.26×	0.2990	0.6547	2.32×	160,148	155,028	1.02×	318,377	1.33e+06	3.25×	2.56	2.49	0	16/1836/160
Sequencing	Exact	no	51	4.51	2.07	0.61×	1.89	1.25	0.74×	125,204	53,422	0.45×	898,317	462,892	0.55×	1,672	1,672	0	0/51/0
	Exact	yes	51	5.76	2.54	0.59×	0.6226	0.6679	1.21×	16,373	16,373	1.00×	125,213	125,213	1.00×	1,672	1,672	0	0/51/0
	Restricted	no	585	8.46	23.61	1.36×	0.9710	2.45	1.64×	76,117	68,647	0.72×	434,831	418,119	0.83×	237,249	230,490	0	290/295/0
	Restricted	yes	585	8.88	24.43	1.37×	0.3266	1.85	3.15×	15,538	40,647	1.79×	76,129	233,148	2.00×	237,249	230,490	0	290/295/0
	RelaxPriority	no	377	1.54	17.80	2.80×	0.5470	19.16	5.10×	13,749	11,253	0.64×	323,290	5.55e+06	3.32×	23,138	23,065	0	21/171/185
	RelaxPriority	yes	386	1.92	29.23	3.28×	0.1828	4.97	5.85×	2,085	6,931	1.97×	37,955	825,239	4.68×	22,927	22,886	0	22/199/165